Capstone Report

Group 3

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Abstract

Multi-agent large language model systems often rely on one model to critique another, but a revised answer alone does not reveal how the receiving model internally processes that feedback. We study feedback uptake in a Solver–Critic system using MuSiQue, a multi-hop question-answering benchmark with known answers. A Gemma 3 4B instruction-tuned Solver first answers each question, a Qwen3 4B instruction-tuned Critic independently reviews that answer, and the Solver then chooses between its original response and the Critic's recommendation. We capture Solver residual-stream activations across nine transformer layers and use sparse autoencoders (SAEs) to represent those activations as sparse features. Linear probes identify features associated with adoption versus retention, while held-out causal interventions test whether altering selected features changes the Solver's decision. The collected candidate-presentation dataset contains 22,355 sampled questions; the current labeling policy identifies 5,834 adoptions, 4,810 retentions, 8,451 agreements, and 3,260 episodes excluded from the primary behavioral analysis. This design separates predictive association from causal evidence and provides a reproducible framework for investigating when language models defer to, resist, or agree with another model's feedback.

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1 Introduction

Large language models (LLMs) are increasingly deployed as components of multi-agent systems in which models critique, debate, verify, or revise one another's outputs. These systems can improve factuality and reasoning by combining multiple model responses [3]. Their success, however, depends on more than whether a Critic can produce useful feedback. A Solver must also decide whether that feedback should change its answer. Blind deference can propagate incorrect recommendations, while indiscriminate resistance can prevent useful correction. Prior work on self-correction and sycophancy shows that externally prompted revision does not necessarily imply reliable error correction [6, 8].

Behavioral evaluation can reveal whether a Solver changed its answer, but it cannot by itself identify the internal computations responsible for that decision. A representation may correlate with feedback adoption because it encodes disagreement, answer content, Critic quality, or a downstream consequence of the decision. Establishing a mechanism therefore requires two steps: finding internal features that predict behavior on held-out data and intervening on those features to test whether they influence behavior.

Sparse autoencoders (SAEs) offer a practical representation for this analysis. They decompose dense neural activations into a larger set of sparsely active features, some of which can be more interpretable than individual model dimensions [2, 4]. We use SAEs to encode the Solver's hidden states while it processes Critic feedback. Regularized linear probes then test whether the resulting feature activations predict whether the Solver adopts the Critic's recommendation or retains its original answer. Finally, selected features are ablated or modified during a replayed forward pass, allowing us to compare targeted interventions with random-feature controls.

Our experiments use MuSiQue, a benchmark designed to require composition over multiple reasoning steps [9]. For each question, a Gemma 3 4B instruction-tuned Solver produces an initial answer and a Qwen3 4B instruction-tuned Critic independently produces a review and recommendation. The Solver is subsequently shown its prior answer and the Critic's candidate answer in randomized order. This candidate-presentation protocol makes adoption and retention more directly observable than unconstrained free-form revision while preserving agreement and unresolved cases as separate outcomes.

This project addresses three primary research questions:

1. Which internal Solver representations predict adoption versus retention of Critic feedback?
2. Do interaction-trained SAEs isolate feedback-related structure more effectively than a general pretrained SAE?
3. Do targeted interventions on probe-selected SAE features causally change feedback-uptake behavior relative to appropriate controls?

The main contribution is an end-to-end, reproducible workflow connecting behavioral data collection, leakage-aware partitioning, sparse representation learning, predictive probing, and causal intervention. The workflow also retains agreement, refusal, ambiguous, and malformed responses rather than forcing every interaction into a binary target. This distinction is important: a model cannot meaningfully adopt a Critic's answer when both agents already agree, and uncertain text matches should not be treated as ground-truth behavior labels.

2 Related Work

2.1 Multi-agent critique and model revision

Multi-agent LLM methods use interaction among model instances to improve a final response. Multi-agent debate, for example, allows agents to exchange arguments and revise their answers over several rounds, producing gains on some factual and reasoning tasks [3]. A Solver–Critic system is a simpler instance of the same idea: one model proposes an answer and a second model evaluates it before revision. These approaches are motivated by the possibility that one agent can expose errors missed by another.

Revision is not automatically equivalent to correction. Huang et al. show that LLMs can struggle to improve reasoning through intrinsic self-correction without reliable external feedback [6]. Conversely, models may accept external signals for reasons unrelated to their correctness. Work on sycophancy documents cases in which models adapt responses to a user's stated view even when doing so reduces truthfulness [8]. Together, these findings motivate separating three questions: whether feedback is correct, whether the Solver adopts it, and whether adoption improves the final answer. Our primary behavioral target concerns adoption rather than correctness, while correctness is retained for stratified analysis.

2.2 Sparse autoencoders for language-model representations

Transformer residual streams are high-dimensional and superposed: a single dimension need not correspond to a single human-interpretable concept. SAEs learn an overcomplete dictionary that reconstructs dense activations from a sparse set of latent features. Cunningham et al. demonstrated that SAE features can be substantially more interpretable than raw neuron activations [2]. Subsequent work studied how SAE design and scale affect sparsity, reconstruction quality, and downstream interpretability [4].

SAE features remain learned model components rather than guaranteed semantic units. Reconstruction error, inactive features, correlated latents, and feature splitting can all complicate interpretation. For that reason, this study compares multiple SAE variants and measures both reconstruction behavior and downstream predictive utility. A general pretrained SAE tests whether broadly learned features transfer to feedback processing, whereas interaction-trained SAEs test whether training on the experiment's activation distribution better captures the relevant structure.

2.3 Probing and causal validation

Linear probes test whether information can be decoded from an intermediate representation [1]. Their simplicity makes coefficients useful for ranking candidate features, but probe performance alone does not show that the model uses those features to produce its behavior. Probes may exploit correlated information or properties of the dataset rather than a causal mechanism. Control tasks, held-out evaluation, and careful capacity selection are therefore necessary when interpreting probe results [5].

Feature intervention provides a stronger test. Work on sparse feature circuits demonstrates how sparse features can support both circuit discovery and targeted model editing [7]. Following this general principle, our analysis uses probe coefficients only to nominate candidate SAE features. We then replay held-out Solver interactions while ablating selected features and compare the resulting adoption rate with no-intervention and random-feature controls. This progression from decoding to intervention is intended to distinguish features that merely describe feedback uptake from features that help produce it.

2.4 Multi-hop question answering as an experimental setting

MuSiQue constructs multi-hop questions by composing connected single-hop questions and filtering shortcuts that allow models to bypass the intended reasoning chain [9]. It is useful for the present study because it supplies question identifiers, supporting passages, known answers, and answer aliases at sufficient scale for repeated Solver–Critic interactions. Its range of two- to four-hop questions also creates variation in difficulty. At the same time, exact or normalized string matching cannot resolve every semantic equivalence. Our labeling procedure therefore uses question-local aliases and conservative text rules, and excludes cases whose adoption status cannot be determined reliably.

3 Exploratory Data Analysis

3.1 Dataset construction

The current candidate-presentation collection contains 22,355 sampled MuSiQue questions. Each episode follows the same sequence: the Solver produces an initial answer (A1), the Critic independently reviews that answer and supplies a recommendation (C1), and the Solver produces a second answer (A2) after both candidates are presented in randomized order. Generation is deterministic for the recorded run. Solver activations are captured at decoder layers 4, 8, 17, 25, 29, 30, 31, 32, and 33 so that later analyses can compare where feedback-related behavior is most decodable.

Questions are assigned to frozen train, validation, test, and intervention partitions by question identifier. All records associated with one question remain in the same partition. The training partition is used to fit SAEs and probes, validation supports model and feature selection, test is reserved for final predictive evaluation, and the intervention partition is reserved for causal experiments. This question-level separation prevents related interactions from appearing on both sides of an evaluation boundary.

3.2 Behavior-label distribution

The labeling policy compares A1, C1, and A2 using normalized text, question-specific answer aliases, and conservative rules for refusals and ambiguous responses. Each episode receives one of four mutually exclusive labels:

- **Adoption**: A1 and C1 differ, and A2 follows the Critic.
- **Retention**: A1 and C1 differ, and A2 retains A1.
- **Agreement**: A1 and C1 agree, and A2 preserves that answer.
- **Out**: the behavior cannot be compared reliably or the episode fails an inclusion rule.

Adoption and retention together form the primary binary cohort of 10,644 episodes, or 47.6% of the full collection. Within this cohort, adoption accounts for 54.8% and retention for 45.2%. This is substantially more balanced than an earlier free-form-feedback labeling approach, which was dominated by adoption. Nevertheless, class-aware metrics such as balanced accuracy, per-class precision and recall, F1 score, and area under the ROC curve remain preferable to raw accuracy.

The label proportions are broadly similar across the four frozen partitions, which reduces the risk that an observed evaluation difference is driven only by a major shift in the outcome distribution. The 23 unassigned questions are quarantined as *Out* and are not used for model fitting or evaluation.

Table 1: Behavior labels in the candidate-presentation collection.

Label	Episodes	Share
Adoption	5,834	26.1%
Retention	4,810	21.5%
Agreement	8,451	37.8%
Out	3,260	14.6%
Total	22,355	100.0%

Table 2: Behavior labels by frozen experimental partition.

Partition	Adoption	Retention	Agreement	Out
Train	4,146	3,465	6,063	2,262
Validation	625	491	901	396
Test	529	416	755	292
Intervention	534	438	732	287
Unassigned	0	0	0	23

3.3 Data-quality observations

Several response patterns require special handling. The collection includes 1,060 episodes in which both A1 and the Critic refuse or reject the premise; these are generally agreements rather than adoption events. Other ambiguous patterns include letter-only candidate selections, citation-only replies, hyphenation differences, hybrid answers that combine both candidates, and Critic responses that repeat A1 while nominally disagreeing. The final rules retain explicit flags for these cases and exclude uncertain comparisons from the primary binary target.

These observations constrain the interpretation of downstream probes. A probe could learn answer length, refusal style, candidate position, or Critic correctness instead of an internal tendency to accept feedback. Downstream analysis should therefore include text-only and majority-class baselines, report results by relevant behavioral flags, and verify that candidate order is not predictive of the label. Probe-selected features should be interpreted as correlates until targeted intervention demonstrates an effect beyond random-feature and reconstruction controls.

4 Methodology

4.1 Behavioral labels

Every episode is assigned one of the four behavior labels defined in Section 3: Adoption, Retention, Agreement, or Out. Only Adoption and Retention enter the binary target used by the probes: Adoption is the positive class and Retention the negative class. The labels come from two sources, which we keep distinct throughout the paper.

Rule-based reference labels. The labels reported in Section 3 were produced by deterministic rules that compare A1, the Critic's recommendation (C1), and A2 using normalized text, answer aliases, and explicit handling of refusals, letter-only replies, citations, and hybrid answers. These rules were audited against 46 team-reviewed episodes and are frozen. We use them as the reference baseline and for every analysis of the original MuSiQue collection.

LLM-derived training labels. For the combined MuSiQue and synthetic collection, the training targets are labels generated by Claude Opus 5, accessed through Amazon Bedrock with a fixed prompt (`claude_binary_v3_candidates`), medium reasoning effort, and an 8,000-token output limit. For each episode the labeler receives the question, A1, the Critic's recommended answer, and A2, presented in the same candidate order the Solver saw. It returns one of the four labels with a one-sentence reason. Episodes whose Critic gave no committed answer are labeled Out by rule. API, parsing, or truncation failures are recorded as technical exclusions and are never counted as a behavioral label. These are model-generated labels, not independently verified ground truth.

Agreement with the reference labels. Before adopting the LLM-derived labels, we compared them with the reference labels on a 250-episode sample from the training partition of the original collection (Table 3). On the Adoption–Retention decision that defines the probe target, the two sources agree in every case (100 of 100 stratified episodes). Almost all disagreement falls in the reference labels' Out class: the LLM labeler assigns a behavioral label to 47 of the 50 episodes that the rules leave out, many of them long-sentence or letter-only replies that the rules exclude conservatively. On the 46 team-reviewed episodes, the LLM labels match the team's judgment in 38 cases, compared with 26 for the rule-based labels. The sample ran with no technical failures, at about 2.3 seconds and $0.006 per episode.

Table 3: LLM-derived labels against the rule-based reference labels on a stratified 200-episode evaluation set (50 per reference label) from the training partition.

Reference label	LLM-derived label			
	Adoption	Retention	Agreement	Out
Adoption	50	0	0	0
Retention	0	50	0	0
Agreement	0	1	49	0
Out	23	11	13	3

Cohen's κ on this set is 0.68, and accuracy re-weighted to the training partition's label mix is 0.87. The lower agreement on Out reflects a difference in how ambiguous replies are treated, not a disagreement about which candidate the Solver chose.

4.2 Aligning labels with activations

Labels are exported together with the frozen partition manifest as a package that the probe and feature-selection code read directly. The export refuses an episode labeled twice, an episode in a labeled partition that has no label, and any label whose partition or activation index disagrees with the collection. It also refuses a label outside the four classes. Technical exclusions are listed with their reasons, so every collected episode is either labeled or explicitly excluded. Partitions that are not labeled stay in the manifest without labels and cannot be read by accident. On the original collection, the export joins 22,130 episodes to their nine-layer activations with every row index verified.

5 Experiments

5.1 From free-form feedback to candidate presentation

The first collection used a free-form protocol: the Solver revised its answer after reading the Critic's full feedback text. The same questions and models were then collected again with the candidate presentation, in which the Solver chooses between its first answer and the Critic's recommendation, shown as two labeled options. We compare the Adoption share of the binary cohort under the two protocols. Because the earlier collection was labeled with an earlier rule set, we report it with both its own rules and the LLM-derived labels.

5.2 Candidate-order control

In the candidate presentation, the Solver sees its own first answer and the Critic's recommendation as two labeled options. Each question is assigned one of two orders before generation, either A1 first or Critic first. The assignment is seeded and balanced within each hop group, so it is independent of the question's content and outcome. We compare the Adoption share of the binary cohort between the two orders in every partition, with 95% Wilson intervals. A difference would show that answer position, rather than the content of the feedback, influences the Solver's choice.

5.3 Synthetic-question pilot

To enlarge the Adoption and Retention cohort, we generated synthetic multi-hop questions in the MuSiQue format with Claude Opus 5. Each question comes with its supporting paragraphs, three generated distractor paragraphs, and filler paragraphs drawn from MuSiQue. Before adding them to the training data, we ran a pilot to check whether they produce feedback behavior comparable to MuSiQue.

The pilot collected 100 synthetic questions with the same models, protocol, candidate presentation, and decoding settings as the MuSiQue collection. As a control, we took 100 MuSiQue questions from the frozen training partition (seed 20261030), matched to the pilot's hop mix: 34 two-hop, 20 *3hop1*, 13 *3hop2*, 10 *4hop1*, 9 *4hop2*, and 14 *4hop3* questions. Both groups were labeled with the same LLM labeler and settings (Section 4.1). We also measured where the two groups differ: A1 and Critic accuracy, whether a lexical retriever ranks the supporting paragraphs above the rest, whether wrong answers come from a non-supporting paragraph, and how often the Critic refuses.

5.4 Paired filler ablation

The pilot suggested that the filler paragraphs, drawn at random from MuSiQue, rarely compete with the evidence. To test this directly, we rebuilt the same 100 synthetic questions with only the filler paragraphs replaced. Each filler is swapped, in place, for a paragraph drawn (with a fixed seed) from the 20 eligible MuSiQue paragraphs most similar to the question and its supporting paragraphs (TF-IDF). Every question, answer, alias, decomposition, supporting paragraph, and generated distractor is unchanged. A filler that mentions any hop's answer or alias is excluded, so a replacement cannot restate or contradict the evidence.

Both conditions are collected in the same environment, in the original question order, so per-question seeds and candidate orders are identical and the fillers are the only difference. The comparison is paired: for each question, we record whether its label changes between the original and the related fillers.

6 Results

6.1 Free-form feedback made the Solver adopt almost everything

Under the free-form protocol, the Solver almost always followed the Critic (Table 4). Retention was too rare to train a probe on: 8% of the binary cohort under the rule-based labels and 3% under the LLM-derived labels. The candidate presentation, with the same questions and models, brought the Adoption share down to 54.8%, which gives a usable binary target. This is why every later collection uses the candidate presentation.

Table 4: Adoption share of the binary cohort under the two Solver protocols. The free-form rows use the training partition (rule-based) or a 1,000-episode sample (LLM-derived); the candidate row uses the full collection.

Protocol	Label source	Binary episodes	Adoption share
Free-form feedback	Rule-based (earlier rules)	4,177	91.7%
Free-form feedback	LLM-derived	586	97.3%
Candidate presentation	Rule-based (final rules)	10,644	54.8%

6.2 Candidate order shifts the Adoption–Retention balance

Presentation order changes the Solver's choice. When the Critic's recommendation is listed first, the Adoption share of the binary cohort is 57.8% (95% CI 56.4–59.1), against 51.9% (50.5–53.2) when A1 is listed first (Table 5). The gap appears in every partition, with the Critic-first share higher by 4 to 9 percentage points (Figure 1).

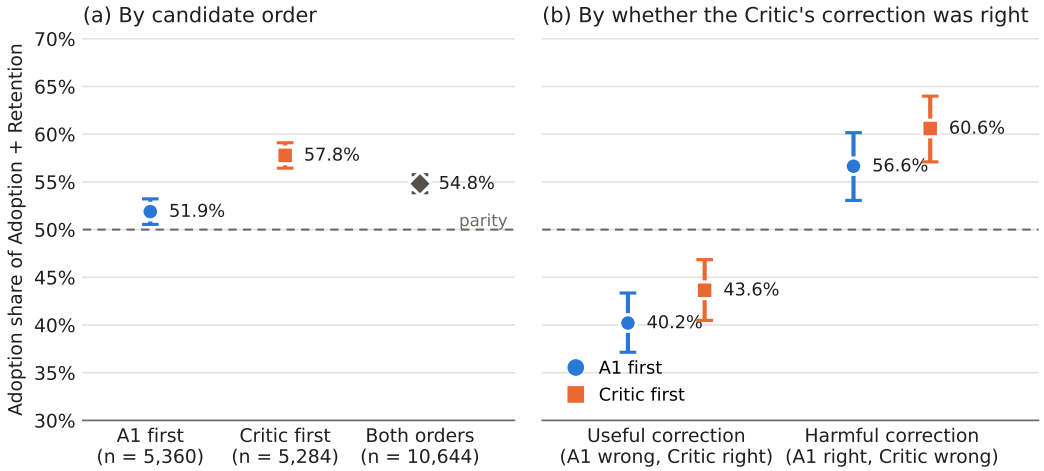


Figure 1: Adoption share of the binary cohort by candidate order, with 95% Wilson intervals (rule-based reference labels).

Because orders are assigned in equal numbers, the effect largely cancels in the overall label mix. It still matters for the probes. A probe could predict Adoption partly from which candidate came first, so candidate order should be reported as a baseline feature and checked as a confound in every probe and feature analysis.

Table 5: Adoption share of the binary cohort (Adoption / (Adoption + Retention)) by candidate order, rule-based reference labels, with 95% Wilson intervals.

Partition	A1 first	Critic first
Train	51.5% (50.0–53.1)	57.4% (55.9–59.0)
Validation	53.5% (49.3–57.5)	58.6% (54.5–62.7)
Test	51.5% (47.0–56.1)	60.2% (55.8–64.4)
Intervention	53.1% (48.7–57.4)	57.0% (52.5–61.4)
All	51.9% (50.5–53.2)	57.8% (56.4–59.1)

6.3 Synthetic questions produce little disagreement

The synthetic questions did not behave like MuSiQue questions (Figure 2, Table 6). Only 16 of 100 synthetic episodes were Adoption or Retention, against 56 of 100 matched MuSiQue episodes, and the two 95% intervals do not overlap. The difference arises before the Solver ever sees the feedback. The Solver answered 78% of synthetic questions correctly on its first attempt, and the Critic 90%, so in most episodes both agreed on the right answer.

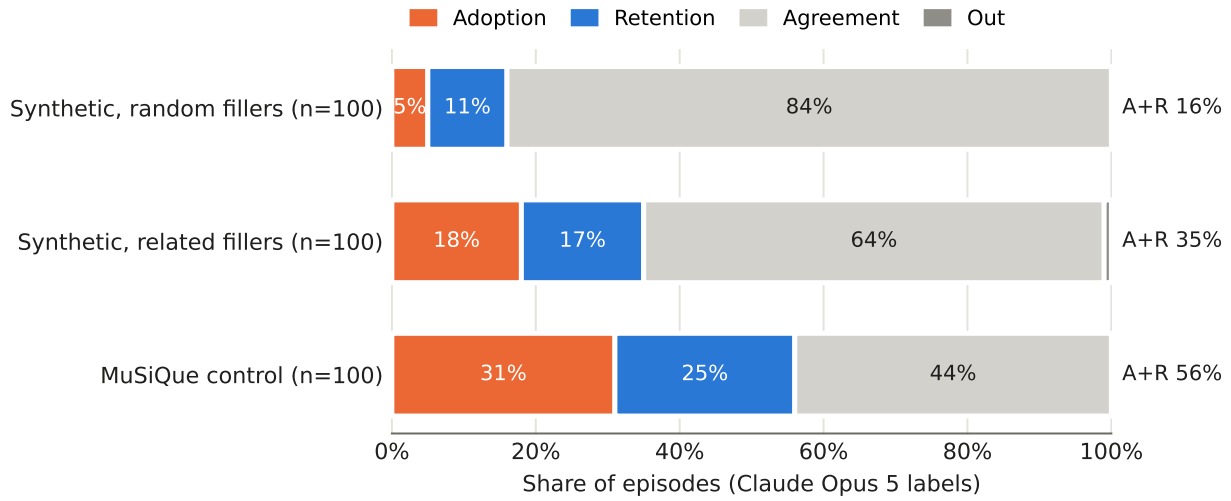


Figure 2: Behavior labels for the synthetic pilot with its original (random) fillers, the same questions with related fillers, and a hop-matched MuSiQue control (LLM-derived labels, $n = 100$ each).

A follow-up analysis of the same episodes points to several causes, none dominant on its own:

- **Weak fillers.** The filler paragraphs rarely compete with the evidence. A synthetic question has 1.5 non-supporting paragraphs on average that share two or more of its content words, against 10.7 for MuSiQue. A lexical retriever ranks every supporting paragraph near the top (within the first hops + 2 positions) for 74% of synthetic questions, against 16% for MuSiQue.
- **Scoring.** Strict string matching undercounts MuSiQue accuracy, because many MuSiQue first answers are full sentences. With lenient matching, A1 accuracy is 83% for synthetic and 35% for MuSiQue, so about 10 of the 57-point gap is a scoring artifact.
- **Defects in MuSiQue.** Part of MuSiQue's disagreement comes from defects in the dataset

Table 6: Synthetic pilot, paired filler ablation, and hop-matched MuSiQue control (LLM-derived labels, $n = 100$ each). One related-filler question was excluded at collection because the Critic gave no recommendation; it is counted as Out, and accuracies use the 99 collected episodes.

	Synthetic		MuSiQue
	Random fillers	Related fillers	
Adoption	5	18	31
Retention	11	17	25
Agreement	84	64	44
Out	0	1	0
Adoption + Retention (95% CI)	16% (10–24)	35% (26–45)	56% (46–65)
Adoption share of the binary cohort	31%	51%	55%
A1 correct	78%	65%	21%
Critic correct	90%	89%	27%

itself: garbled composed questions, gold answers that do not fit the question, and paragraphs that do not state the cited relation. The Critic refused to answer 17% of the control questions and none of the synthetic ones.

MuSiQue's Adoption + Retention rate is therefore not the right target for the synthetic data. Most synthetic disagreements are the clean case, a wrong A1 that the Critic corrects (12 of 16), whereas more than half of MuSiQue's disagreements involve two wrong answers (32 of 56). The goal is enough clean disagreement, not a match to MuSiQue's rate.

6.4 Related fillers

Replacing only the filler paragraphs more than doubled the useful disagreement: Adoption + Retention rose from 16% to 35% of episodes (Table 6). Pairing the same questions across the two conditions, 22 episodes became Adoption or Retention and 3 stopped being so (exact McNemar test, $p < 0.001$). The Adoption share of the binary cohort also moved from 31% to 51%, close to MuSiQue's 55%.

The related fillers act mainly on the Solver's first answer. A1 accuracy fell from 78% to 65%, while the Critic stayed near 90%. In 17 of the 22 new disagreements the Solver was wrong and the Critic right, which is the clean case of correctable feedback. The new wrong answers are genuine reasoning errors rather than noise: of 19, twelve take an intermediate entity or the wrong span from a supporting paragraph, four take an answer from a related filler, two from a generated distractor, and one appears nowhere in the context.

Two checks support attributing the change to the fillers. Collecting the original questions again in the same environment reproduced every A1, Critic answer, and A2 of the original pilot exactly. On the 57 episodes whose three answers did not change, the labeler assigned the same label in every case.

7 Discussion

7.1 Limits of the behavioral labels

The probe targets for the combined collection are generated by a language model. On the Adoption–Retention decision they agree with the frozen rule-based labels in every sampled case, and on the team-reviewed episodes they match the team's judgment more often than the rules do. They remain model judgments, however, so we report them as LLM-derived labels rather than ground truth. The two sources differ mainly on ambiguous replies that the rules place in Out. A probe result that holds under one label source but not the other should be read with that difference in mind, and the frozen rule-based labels remain available for checking any headline result on the original collection.

Both label sources depend on the Critic's recommended answer, the only feedback the Solver sees in the candidate presentation. Neither judges whether an answer is factually correct, so the binary target measures uptake of feedback, not improvement in accuracy.

7.2 Limits of the synthetic pilot

The pilot has 100 questions per group, which is enough to detect the large difference in disagreement but leaves wide intervals for any single hop type. Its synthetic questions use fictional entities, so the models cannot draw on prior knowledge in either direction. This makes them a cleaner test of reading the passages, but less like real questions. The comparison with MuSiQue is also affected by MuSiQue's own defects and by strict answer matching, both of which inflate MuSiQue's apparent difficulty. We therefore treat the pilot as evidence about which generation choices make questions too easy, not as a measure of how much harder synthetic questions should be.

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